High-Power LED Stroboscope



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Stroboscope is a convenient and accurate enough instrument to measure speeds of rotating objects in homes or industries. One can use it to find the speed of fans, motors or any other rotating object.

Stroboscope is nothing but a flashing light that can provide sharp light pulses at a variable rate. If a rotating object is observed in a powerful beam of pulsed light with frequency matching the rotations per second of the rotating object, the rotating object appears to be stationary. So the speed of any rotating object can be calculated by varying the pulse rate until the rotating object appears stationary. Once this state is achieved, revolutions per minute (rpm) of the rotating object will be equal to the pulse time.

Circuit and working

Fig. 1 shows the circuit of microcontroller-based stroboscope. It comprises microcontroller PIC16F73 (IC1), regulator 7805 (IC2), three-digit common anode 7-segment display KLT363 (DIS1) and a few discrete components.

Microcontroller. PIC16F73 (IC1) is the heart of the stroboscope and provides a wide range of short pulses. It is a powerful microcontroller that provides an ideal solution for hobby and industrial development.

PIC16F73 is an 8-bit, high-performance, low-power RISC CPU. Its main features are 4kB flash, 192 bytes of RAM, three input/output (I/O) ports, 8-bit 5-channel analogue-to-digital converter (ADC), three timers and watchdog timer with its own on-chip R-C oscillator for reliable operation. The microcontroller can recognise and execute only 35 simple instructions. All instructions, except branches, are single-cycle.

Port pins RB0 through RB7 of microcontroller IC1 are connected to segments 'a' through 'g' and 'dp' of three-digit 7-segment display DIS1 as shown in Fig. 1.

Port pins RC1, RC2 and RC3 are connected to the bases of transistors T4, T3 and T2 to drive common-anode pins 12, 9 and 8 of DIS1, respectively. When these port pins go low, transistor T2 is driven into saturation, providing supply to common-anode pins of DIS1.

Microcontroller IC1 provides segment data and display-enable signals simultaneously in time-division multiplexed mode for displaying a particular number on the 7-segment display unit. Segment data and display-enable pulses for the display are refreshed very fast. Thus the display appears to be continuous even though its segments light up one by one.

Multi-turn trimpot VR1 is used to change the time period of stroboscope pulse. Port pin RC0 of microcontroller IC1 drives MOSFET T5 to produce pulsed light through LED2 for speed measurement. Resistor R11 limits the current through LED2. Its value depends on the LED used.

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Low when reset
TP3	Flashing light pulse
TP4	Square wave 50 Hz
TP5	ADC input

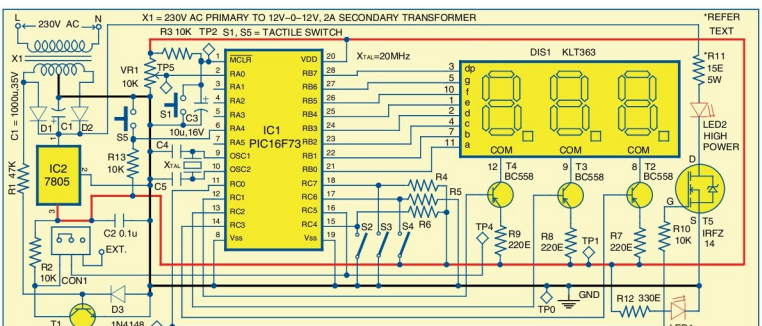


Fig. 1: Circuit of the LED stroboscope